

01

• **RATE OF A CHEMICAL REACTION**
decrease in conc of reactant OR increase in conc of product
Time taken Time taken

• **AVERAGE RATE**
Consider a reaction: $A + B \rightarrow C + D$
 $\frac{\Delta[A]}{\Delta t}, \frac{\Delta[B]}{\Delta t}, \frac{\Delta[C]}{\Delta t}, \frac{\Delta[D]}{\Delta t}$

• **INSTANTANEOUS RATE**
Consider a reaction: $aA + bB \rightarrow cC + dD$
 $-\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$
Unit of Rate = $\text{mol litre}^{-1} \text{s}^{-1}$

Q. During the decomposition of H_2O_2 , 48 g O_2 is formed per minute at a certain point of time. The rate of formation of water at this point is
(a) $0.75 \text{ mol min}^{-1}$ (b) 1.5 mol min^{-1}
(c) $2.25 \text{ mol min}^{-1}$ (d) 3.0 mol min^{-1}

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• **FACTORS INFLUENCING RATE OF REACTION**

Factors	Effect on reaction rate
Increase in concentration	Increases
Increase in temperature	Increases
Presence of catalyst	Increases

Q. Which of the following will lead to an increase in the rate of the reaction?
(a) Decrease in temperature
(b) Decreasing concentration of reactants
(c) Addition of catalyst
(d) Addition of inhibitor

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• **RATE LAW**
Consider a general reaction,
 $aA + bB \rightarrow \text{product}$
Rate = $k[A]^x[B]^y$ (law of mass action)
Rate = $k[A]^x[B]^y$ (rate law expression)
 x & y are determined experimentally and may or may not be equal to a & b
 x & y represents the order of reaction with respect to A & B

• **RATE CONSTANT**
• Larger the value of k , faster is the reaction.
• The value of k changes only with temperature for given reaction.
Unit of rate constant = $(\text{mol})^{1-n} \text{L}^n \text{s}^{-1}$

Q. The rate constant of a zero-order reactions has the unit
(a) s^{-1} (b) $\text{mol L}^{-1} \text{s}^{-1}$
(c) $\text{L}^2 \text{mol}^{-2} \text{s}^{-1}$ (d) $\text{L mol}^{-1} \text{s}^{-1}$

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• **ORDER AND MOLECULARITY**
Consider a general reaction,
 $aA + bB \rightarrow \text{product}$
Rate = $k[A]^x[B]^y$
molecularity = $a + b$
order = $x + y$

Molecularity	Order
Theoretical concept. It cannot be zero, fractional, infinite and imaginary.	An experimentally determined quantity. It can be equal to zero, positive, negative and fractional.

Q. When the rate of the reaction is equal to the rate constant, the order of the reaction is
(a) zero order
(b) first order
(c) second order
(d) third order

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• **PSEUDO ORDER REACTIONS**
Consider the reaction
 $\text{C}_2\text{H}_5\text{Br} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{C}_2\text{H}_5\text{OH} + \text{H}^+$
In these reactions, concentration of water (one of the reactants) is in excess and its concentration remains constant throughout the reaction.
Thus, rate $\propto [\text{C}_2\text{H}_5\text{Br}]$
Therefore, order = 1

Q. For a pseudo first-order reaction, what is the unit of the rate of the reaction?
(a) s^{-1}
(b) $\text{mol L}^{-1} \text{s}^{-1}$
(c) $\text{mol}^{-1} \text{L s}^{-1}$
(d) $\text{mol}^2 \text{L}^2 \text{s}^{-1}$

• ELEMENTARY & COMPLEX REACTIONS

• Reactions occurring only in one step are called elementary reactions while that involving a sequence of elementary reactions, are called complex reactions.

• In case of complex reactions, the slowest step is called rate determining step.

Consider the reaction
 $2O_3 \rightarrow 2O_2$
 Step - 1
 $O_3 + O_3 \rightarrow O_2 + O_2$ (fast)
 Step - 2
 $O + O_3 \rightarrow 2O_2$ (slow)

From slow step
 $r = k [O] [O_3]$
 Here $[O]$
 $[O] = \frac{[O_3]}{[O_2]}$: From fast step
 $r = k [O_3]^2 [O_2]^{-1}$

Q. Suppose the reaction: $A + 2B \rightarrow AB_2$ occurs by the following mechanism:

Step 1: $A + B \rightarrow AB$ slow

Step 2: $AB + B \rightarrow AB_2$ fast

Overall $A + 2B \rightarrow AB_2$

(a) $k[A]$ (b) $k[B]$ (c) $k[A][B]$ (d) $k[B]^2$

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• INTEGRATED RATE EQUATIONS

Zero order
 $k = \frac{[A]_0 - [A]_t}{t}$

First order
 $k = \frac{2.303 \log \frac{[A]_0}{[A]_t}}{t}$
 Second order
 $k = \frac{1}{t} \left[\frac{1}{[A]_t} - \frac{1}{[A]_0} \right]$

Q. A first order reaction has a specific reaction rate of 10^{-2} sec^{-1} . How much time will it take for 20 g of the reactant to reduce to 5 g?

(a) 138.6 sec (b) 346.5 sec
 (c) 693.0 sec (d) 238.6 sec

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• HALF LIFE PERIOD

Zero order
 $t_{1/2} = \frac{[A]_0}{2k}$

First order
 $t_{1/2} = \frac{0.693}{k}$
 Second order
 $t_{1/2} = \frac{1}{k[A]_0}$

FIRST ORDER TRICKS

$t_{75\%} = 2t_{1/2}$

$t_{99\%} = \frac{2.303}{k}$

$t_{99.9\%} = 10t_{1/2}$

Q. The half-life period of zero order reaction is directly proportional to the _____

(a) Rate constant
 (b) Initial concentration of reactants
 (c) Final concentration of reactants
 (d) Concentration of products

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• GRAPHICAL REPRESENTATION

1. Concentration-time graph

a. Zero order b. First order

c. Zero order b. First order

c. Second order

2. Half life - concentration graph

a. Zero order b. First order

c. Zero order b. First order

c. Second order

Q. The graph of $t_{1/2}$ versus initial concentration 'a' is for

(a) First order
 (b) Second order
 (c) Zero order
 (d) Can't predict

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• ARRHENIUS EQUATION

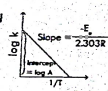
$k = Ae^{-E_a/RT}$

$\log k = \log A - \frac{E_a}{2.303RT}$

NOTE

• For every 10° rise in temperature, rate becomes double and hence, rate constant becomes double.

• A reaction with higher value of E_a will have smaller value of rate constant.



Q. The slope of Arrhenius plot ($\ln k$ vs $1/T$) of first order reaction is $-5 \times 10^3 \text{ K}$. The value of E_a of the reaction is [Given: $R=8.314 \text{ J K}^{-1} \text{ mol}^{-1}$]

(a) -83 kJ mol^{-1} (b) 41.5 kJ mol^{-1}
 (c) 83 kJ mol^{-1} (d) 166 kJ mol^{-1}

CHEMICAL KINETICS

02

OXYGEN FAMILY

PHYSICAL PROPERTIES

Electron affinity
 $-S > Se > Te > Po > O$

CHEMICAL PROPERTIES

Hydrides

Bond Angle
 $H_2O > H_2S > H_2Se > H_2Te$

Thermal Stability
 $H_2O > H_2S > H_2Se > H_2Te$

Acidic Character
 $H_2Te > H_2Se > H_2S > H_2O$

Reducing Power
 $H_2Te > H_2Se > H_2S > H_2O$

B.P
 $H_2O > H_2Te > H_2Se > H_2S$

OXOACIDS OF SULPHUR

1) Thionous type

i) Dithionous acid
 $H_2S_2O_4$
 $(S_2O_4^{2-})$



ii) Dithionic acid
 $H_2S_2O_6$
 $(S_2O_6^{2-})$



iii) Polythionic acid
 $H_2S_nO_6$
 $(S_nO_6^{2-})$



iv) Thiosulphuric acid
 $H_2S_2O_3$
 $(S_2O_3^{2-})$



2) Sulphurous type

i) Sulphurous acid
 H_2SO_3

ii) Sulphuric acid
 H_2SO_4

iii) Pyrosulphuric acid
 $H_2S_2O_7$ (Oleum)

3) Peroxo-type

i) Peroxomonosulphuric acid
 H_2SO_5
 (Cane's acid)



ii) Peroxodisulphuric acid
 $H_2S_2O_8$
 (Marshall's acid)



H₂SO₄

Preparation

Contact process



Properties

3S + 2H₂SO₄ → 3SO₂ + 2H₂O



SO₂

- Act as reducing agent (in aqueous medium)



- Act as oxidizing agent in the presence of



At elevated temperature (1000K) S₂ is dominant species and is poorly oxidized. S₂ molecule which has two unpaired electrons in π orbitals like O₂.

All these compounds are Oxyanions occur in the atmosphere.

Physical properties

- Atomic radii → down the group

- IE → down the group

- He is having maximum ionization energy of all known substances

- Ne is having high positive value of electron gain enthalpy

- Electronegativity → down the group

- He is having highest electronegativity among all known substances (2.4)

COMPOUNDS OF FLUORINE

Xenon Fluoride compounds



d-Block transition elements (group 3-10)

Electronic configuration: $(n-1)d^{1-10} ns^2$

Exceptions: $Cr: 4s^1 3d^5$ $Cu: 4s^1 3d^{10}$ $Pd: 5s^0 4d^{10}$ Non-typical transition elements: Zn, Cd, Hg

Physical properties:

High melting and boiling point

Melting point: s-block metals < d-block metals

$Sc < Ti < V < Cr < Mn < Fe < Co < Ni < Cu < Zn$ (3d series)

Melting point: $Zn > Cd > Hg$

$Cu > Ag > Au$

High enthalpies of atomization (highest for V in 3d series)

Atomic Radius:

3d series: $Sc > Ti > V > Cr > Mn > Fe > Co > Ni < Cu < Zn$

In a group 3d < 4d < 5d (Lanthanide contraction)

eg: $Ti < Zr < Hf$

Smallest radius - Ni

Largest radius - La

Density:

s-block < d-block

3d series: $Sc < Ti < V < Cr < Mn < Fe < Co < Ni < Cu < Zn$

In a group 3d < 4d < 5d

Ionization enthalpy: increases from left to right

Oxidation states: Variable; higher O.S. stable down the group

Trends in E° : E° for Mn, Ni and Zn are more negative than expected.

Trends in E° : variable.

Chemical reactivity and E° values: Variable Ti^{2+} , V^{2+} and Cr^{2+} are strong reducing agents.

Magnetic properties: Diamagnetism and paramagnetism.

Formation of colored ions: due to d-d transitions

Form a large number of complex compounds

Forms interstitial compounds: Non-stoichiometric and are neither ionic nor covalent.

Alloy formation: Due to similar atomic sizes (15% difference in metallic radius)

Lanthanoids

Electronic configuration: $4f^{1-14} 5d^{0-1} 6s^2$ (6d: 4f 5d 6s²)

Atomic and ionic sizes: Decreases from La to Lu (Eu is the largest)

Oxidation states: Most common is +3.

Some elements exhibit +2 and +4.

General characteristics:

Silvery white soft metals and tarnish rapidly in air.

Hardness increases with increasing atomic number.

Metallic structure and good conductors of heat and electricity.

Variable density

Trivalent Lanthanoid ions are coloured.

Ionisation Enthalpies: Low third ionisation enthalpies.

Good reducing agents.

MISCH METAL - Alloy of Ln (95%), Fe (5%) & S, C, Ca, Al etc..

Uses

- In production of iron and steels.
- TiO in pigment industry.
- MnO₂ in dry battery cells.
- As catalysts in industry.
- Ni complexes: polymerization of alkynes and other organic compounds
- AgBr in photographic industry.

Catalysts

Contact process = V_2O_5

Haber process = $Fe^{2+}/Al_2O_3 + K_2O$

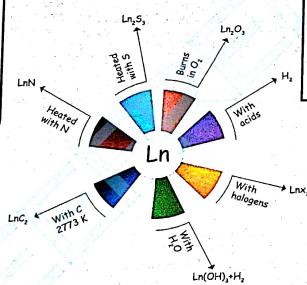
Decomposition of $KClO_3$ = MnO_2

Ostwald process = Pt/Rh

Zeigler Natta catalyst = $TiCl_4 + (C^*_2H_5)_3Al$

Hydrogenation of Alkene = Ni/Pd

Wilkinson's catalyst = $RhCl(PPh_3)_3$



Compounds of d-block elements

Potassium dichromate $K_2Cr_2O_7$

Preparation: $4FeCr_2O_4 + 8Na_2CO_3 + 7O_2 \rightarrow 8Na_2Cr_2O_7 + 2Fe_2O_3 + 8CO_2$
 $2Na_2Cr_2O_7 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$
 $Na_2Cr_2O_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$

Properties: Strong oxidising agent

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

$Cr_2O_7^{2-}$ (Orange)

Cr^{3+} (Green)

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Potassium permanganate $KMnO_4$

Preparation: $2MnO_2 + 4KOH + O_2 \rightarrow 2KMnO_4 + 2H_2O$
 $3MnO_2 + 4H^+ \rightarrow 2MnO_4^{2-} + MnO_2 + 2H_2O$
 $2MnO_4^{2-} + 5S_2O_8^{2-} + 8H^+ \rightarrow 2MnO_4^- + 10SO_4^{2-} + 16H^+$

Intense colour

Paramagnetism

Strong oxidising agent

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

MnO_4^- (Purple)

MnO_4^- (Purple)

MnO_4^- (Purple)

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MnO_4^- (Purple)

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MnO_4^- (Purple)

MnO_4

MONODENTATE LIGAND		BIDENTATE LIGAND	CHELATE LIGAND	GEOMETRICAL ISOMERISM	OPTICAL ISOMERISM
Anionic	X^- - Halido OH^- - Hydroxo NO_2^- - Nitrito SO_4^{2-} - Sulphato O^{2-} - Oxo (Oxido) S^{2-} - Sulphido	1) Oxalato $\begin{array}{c} \text{COO}^- \\ \\ \text{COO}^- \end{array}$ 2) Ethane-1,2-diamine [en] $H_2N-CH_2-CH_2-NH_2$	1) EDTA $^{4-}$, en, ox 2) greater Stability	1) $[Ma_2b_2]$ - 2 (cis + trans) 2) $[Ma_2bc]$ - 2 (cis + trans) 3) $[Mabcd]$ - 3 4) $[Ma_2b_2]$ / $[Ma_2bc]$ - 2 (cis + trans) 5) $[Ma_2b_2c]$ - fac & mer 6) $[Ma_2b_2c_2]$ - 5 7) $[Mabcdef]$ - 15 8) $[M(en)_2b_2]$ or $[M(en)_2bc]$ - 2	OPTICAL ISOMERISM ① $M(AA)_3$ 2 Optical isomers (d cis + l cis) ② $M(AA)_2B_2$ Total 3 Stereoisomer (d cis + l cis + trans) CFT SPECTROCHEMICAL SERIES $I^- < Br^- < SCN^- < Cl^- < F^- < C_2O_4^{2-} < H_2O < NCS^- < EDTA^{4-} < NH_3 < en < NO_2^- < CN^- < CO$
Neutral	H_2O - Aqua NO - Nitrosyl NH_3 - Ammine CO - Carbonyl	POLYDENTATE LIGAND 1) Trien, $N(CH_2CH_2NH_2)_3$ - Tridentate 2) Ethylene Diamine Tetra Acetate [EDTA] $\begin{array}{c} \text{OOCCH}_2 \\ \\ \text{N}-CH_2-CH_2-N \\ \quad \quad \\ \text{OOCCH}_2 \quad \text{CH}_2\text{COO}^- \end{array}$ Donor - 4O Atoms 2N Atoms AMBIDENTATE LIGAND 1) NO_2^- 2) SCN^-	WERNER'S THEORY 1) 1 st Valency - O.N 2) 2 nd Valency - C.N $[Pt(NH_3)_5Cl]Cl_3$ 1) 1 st Valency - O.N = 4 2) 2 nd Valency - C.N = 6 3) AgCl Formed per mole of complex $\rightarrow 3$ 4) Total ions per mole of complex - 4		$[ML_4]$ $\begin{cases} dsp^2 \rightarrow \text{Square planar} \\ sp^3 \rightarrow \text{Tetrahedral} \end{cases}$ $[ML_5]$ $\begin{cases} dsp^2 \rightarrow \text{Trigonal bipyramidal} \\ sp^3d \rightarrow \text{Octahedral} \end{cases}$ $[ML_6]$ $\begin{cases} sp^3d^2 \rightarrow \text{Octahedral} \end{cases}$
Cationic	NO_2^+ - Nitronium NO^+ - Nitrosonium				

COORDINATION COMPOUNDS

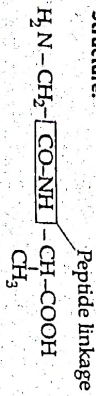
UNPAIRED ELECTRONS - PARAMAGNETIC PAIRED ELECTRONS - DIAMAGNETIC	CFT	CFSE	OMC	APPLICATION OF ORGANO METALLIC COMPOUNDS
$\mu_B = \sqrt{n(n+2)}$ $1 = 1.73 \text{ BM}$ $2 = 2.84 \text{ BM}$ $3 = 3.87 \text{ BM}$ $4 = 4.90 \text{ BM}$ $5 = 5.92 \text{ BM}$ $[NiCl_4]^{2-}$ - sp^3 , Tetrahedral $n=2$ $[NiCO_4]$ - sp^3 , Tetrahedral $n=0$ $[Ni(CN)_4]^{2-}$ - dsp^2 , Sq. planar $n=0$ H_2O act as SFL with Co^{3+} Cr^{3+} - $[t_2g^3 e_g^0]$ forms d^3sp^3 4d and 5d metals with all ligands act as SFL $[M(CO)_6]^{2+}$, $[M(CO)_5]^{+}$, $[M(CO)_4]^{2+}$ + dsp^3 (diamagnetic) $C_2O_4^{2-}$ act as SFL with Co^{3+} F^- act as SFL with Ni^{2+}	CFT 1) Octahedral - Ligand approaches along the axis ($e_g > t_2g$) Average energy of d-orbitals Splitting of d-orbitals in octahedral crystal field. $CFSE = a_1(-0.4\Delta_o) + b_1(0.6\Delta_o) + np$ 2) Tetrahedral - Ligand approaches b/w the axis ($t_2 > e$) Average energy of d-orbitals Splitting of d-orbitals in tetrahedral crystal field.	$CFSE = a_1(-0.6\Delta_o) + b_1(0.4\Delta_o) + np$ here a, b are number of electrons in low energy and high energy d-orbitals HIGH SPIN COMPLEX $\Delta_o < \text{Pairing Energy}$ LOW SPIN COMPLEX $\Delta_o > \text{Pairing Energy}$ For calculation of n, Crystal field stabilisation energy consider newly formed pairing only CFSE 1) Stability $\propto CFSE$ 2) $\Delta_o > \Delta_1$, $\Delta_1 = \frac{4}{9}\Delta_o$ 3) $[Fe(CN)_6]^{3-} < [Fe(CO)_6]$ 4) Jahn Teller distortion by - d^9, d^7, d^9 5) Dissociation constant $= \frac{1}{K_d}$ (K_d is stability constant) 6) Color of d-d transition, $E \propto \frac{1}{\lambda}$ $[Co(en)_3]^{3+} < [Co(NH_3)_6]^{3+} < [Co(H_2O)_6]^{3+}$ Et λ_1 (E = energy of d-d transition) Et λ_1	OMC 1) σ bonded - $RMgX$ (Grignard reagent) - R_2SiCl_2 2) π bonded - Ferrocene 3) σ & π bonded - Metal Carbonyls Mononuclear - $[Fe(CO)_4]$ $[Mn(CO)_5]^{+} < [Cr(CO)_6] < [V(CO)_6]^{-}$ $BO \uparrow BL \downarrow$ $BO \downarrow BL \uparrow$ (w.r to C-O bond) Synergic Bonding	APPLICATION OF ORGANO METALLIC COMPOUNDS 1) Ziegler Natta Catalyst is used in polymerisation of alkene $TiCl_4 + AlEt_3$ 2) Wilkinson's Catalyst is used in hydrogenation $[(C_6H_5)_3P]_3RhCl$ APPLICATION OF COORDINATION COMPOUNDS 1) Cisplatin ($[Pt(NH_3)_2Cl_2]$) is used as an anticancerous agent 2) EDTA is used in Lead poisoning 3) Copper and Iron poisoning - penicillamine & deferrioxamine 4) Vitamin B-12 - Cobalt 5) Chlorophyll - Magnesium 6) Carboxy peptidase - Zinc

(Polymers of α - amino acids)
Amino acids contain $-NH_2$ and $-COOH$ groups.

Classification:

- On the basis of relative number of $-NH_2$ and $-COOH$ group
 - (i) Neutral : equal number of $-NH_2$ and $-COOH$ group
 - (ii) Basic : more number of $-NH_2$ than $-COOH$ group
 - (iii) Acidic : more number of $-COOH$ than $-NH_2$ group
- On the basis of place of synthesis
 - (i) Essential : cannot be synthesized in the body.
 - (ii) Non-essential : synthesized in the body.
- On the basis of shape
 - (i) Fibrous : fibre-like structure
 - (ii) Globular : spherical

Structure:



Denaturation of proteins:

When a protein in its native form is subjected to physical change, globules unfold, helix gets uncoiled and protein loses its biological activity

Two types: Deoxyribonucleic acid (DNA), ribonucleic acid (RNA)

Deoxyribonucleic acid (DNA)	Ribonucleic acid (RNA)
Sugar is β -D-2-deoxyribose	Sugar is β -D-ribose
DNA Contains A=T, G=C	RNA Contains A=U, G=C

Biological functions:

- Chemical basis of heredity.
- Responsible for identity of different species of organisms.
- Nucleic acids are responsible for protein synthesis in cell.

BIO MOLECULES

proteins

enzymes

carbohydrates

nucleic acids

vitamins

Organic compounds required in diet in small amounts to perform specific biological functions for maintenance and growth

Classification:

- (i) Fat soluble : Soluble in fats and oils but insoluble in water. (Vitamins A, D, E and K)
- (ii) Water soluble : B group and vitamin C are soluble in water

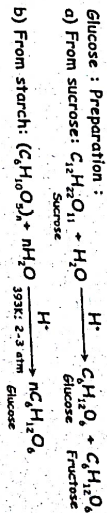
Globular proteins specific for particular reaction and for particular substrate.
Mechanism : Reduces the magnitude of activation energy

Optically active polyhydroxy aldehydes or ketones or compounds which produce such units on hydrolysis

Classification:

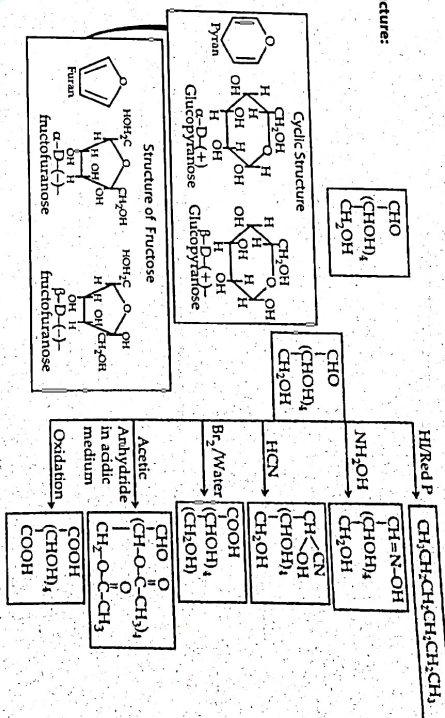
(i) Monosaccharides : (Aldehyde group - aldose, keto group - ketose)

Glucose : Preparation :



b) From starch: $(C_6H_{10}O_5)_n + nH_2O \xrightarrow[393K, 2-3 \text{ atm}]{H^+} nC_6H_{12}O_6$

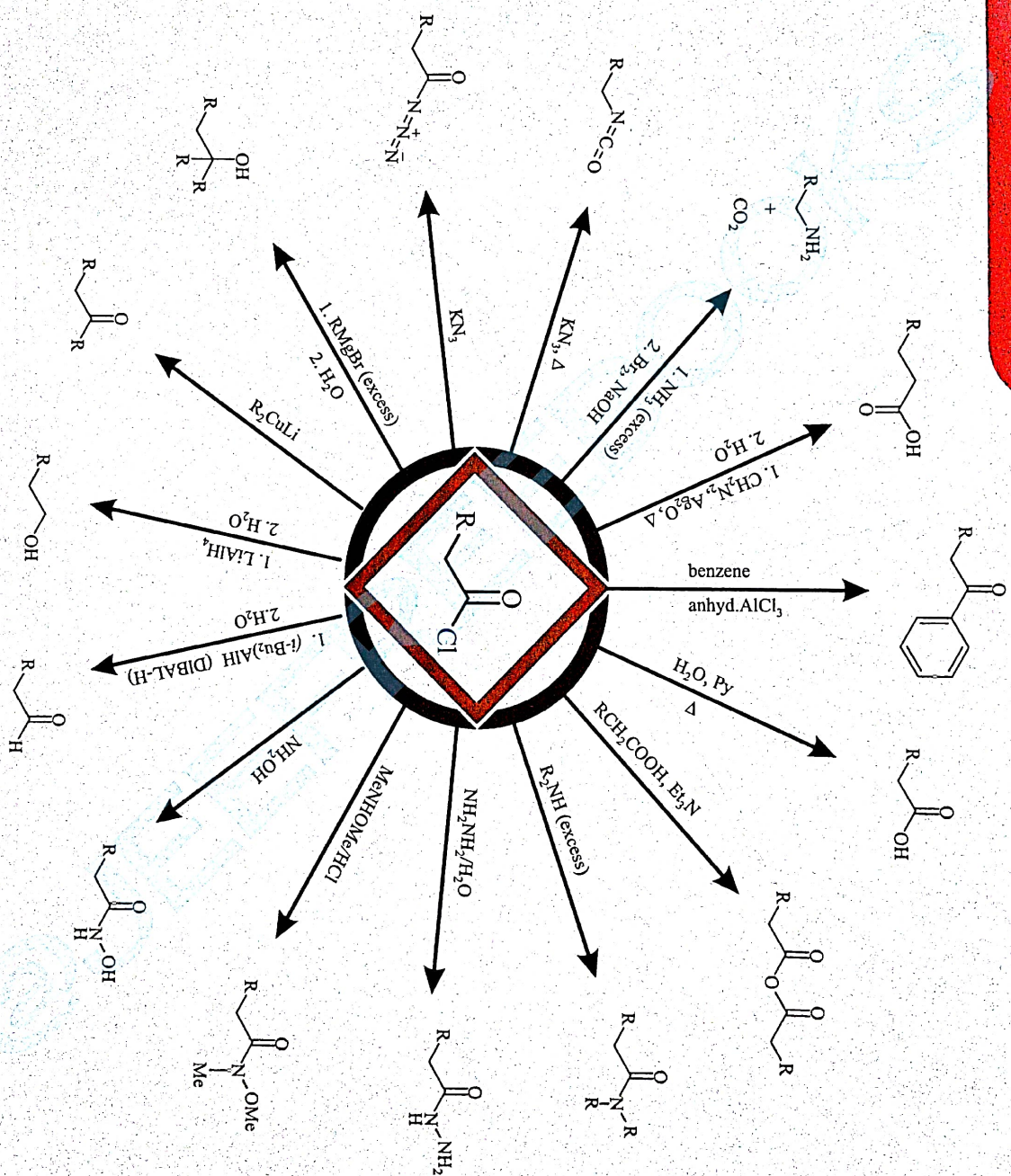
Structure:



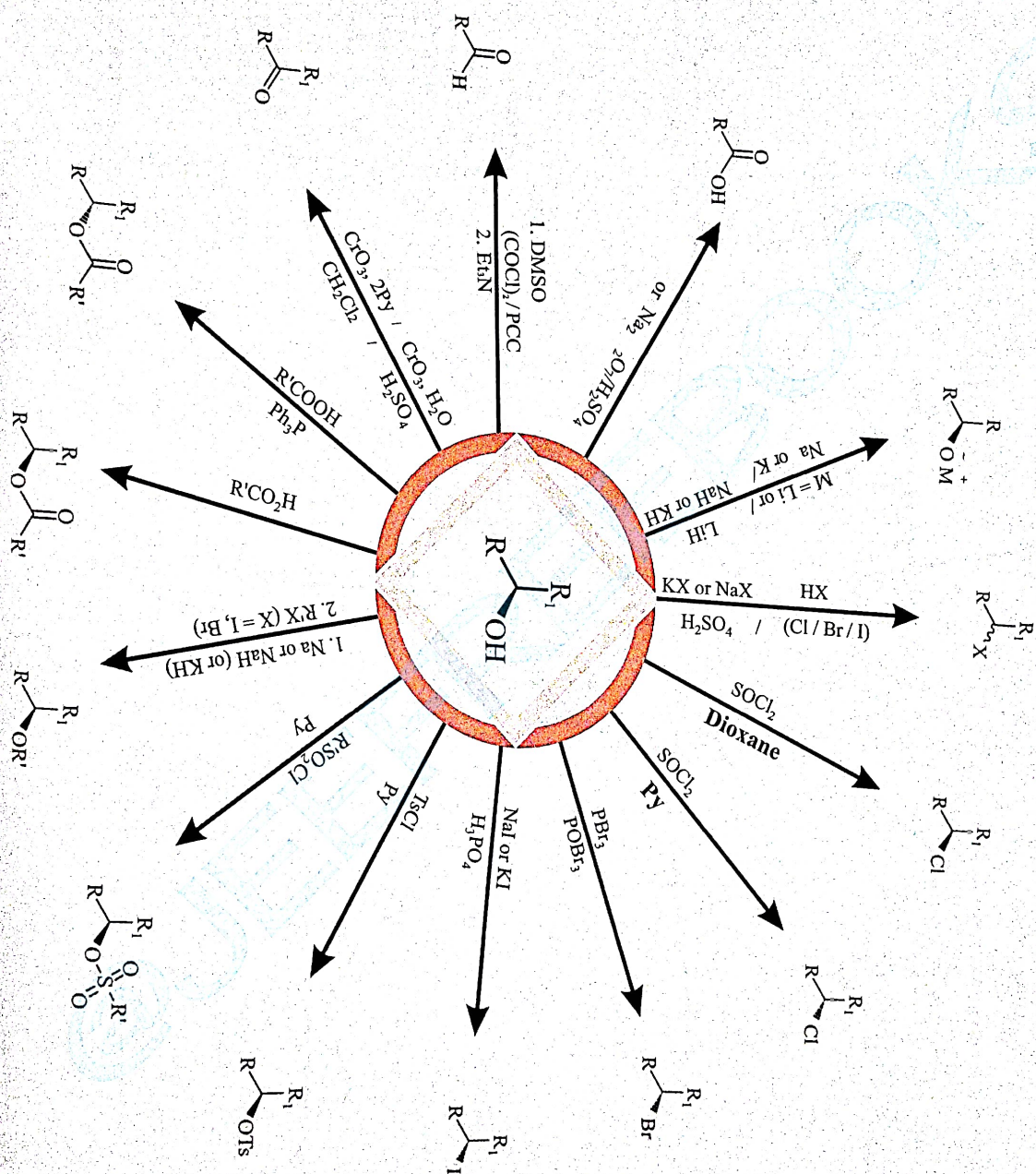
(ii) Disaccharides : Linkage between 2 monosaccharides- Glycosidic linkage

- 1) Sucrose-C-1 of α -glucose and C-2 of β -fructose
- 2) Maltose-C-1 of α -D-glucose with C-4 of another α -D-glucose
- 3) Lactose-C-4 of β -D-glucose with C-1 of β -D-galactose

ACYL (ACID) CHLORIDE



ALCOHOLS



AMINE

